

AQUA Sphere OS

Master Notebook — Roles, Permissions & Business Logic

Complete, corrected transcription of all manager planning notes — written for AI-assisted ("vibe coding") development

How to Read This Notebook

- This single document replaces and combines everything from the manager's three handwritten note sessions — nothing from any session is left out.
- Where the handwriting was unclear, abbreviated, or possibly a slip of the pen, it has been written out in full, plain sentences — every place this happened is marked with a colored Correction or Please Confirm box so nothing gets silently changed.
- Boxes and arrows recreate the manager's original diagrams digitally, in the same shape, so the flow is easy to follow at a glance.
- This is written in plain, simple language on purpose — it's meant to be handed directly to an AI coding tool as a build reference.

Colour key:

Note: *extra context or a business-logic explanation.*

Correction: *the handwriting said one thing but almost certainly meant another — fixed here, flagged so it can be double-checked.*

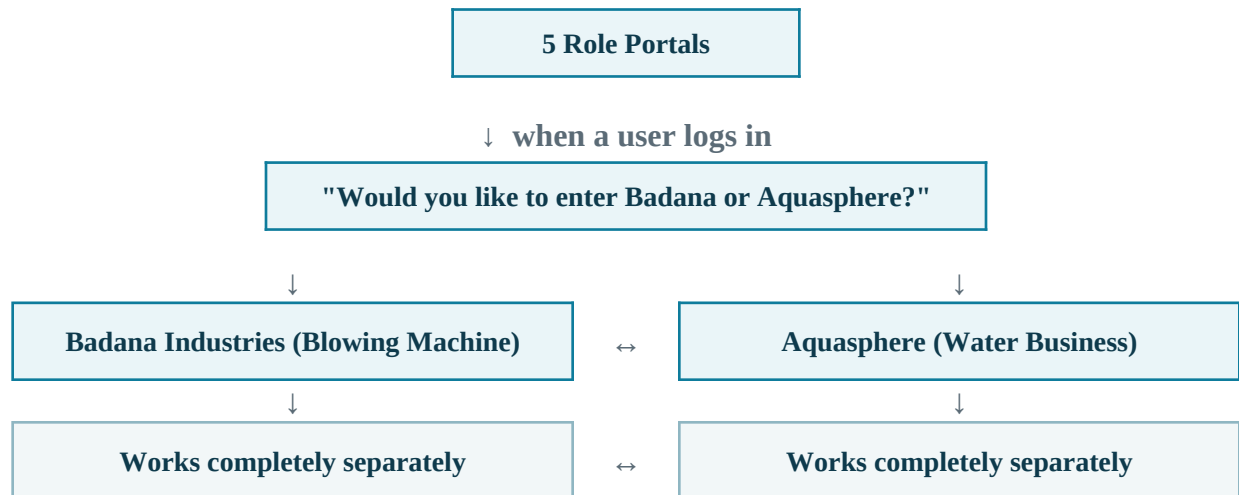
Please confirm with manager: *something written down that is unusual, incomplete, or worth a second look before building it.*

0. System Overview — Two Businesses, One Login

AQUA Sphere actually runs two connected businesses, and the system needs to support both:

- Aquasphere (the water business) — 19L reusable bottles and 0.5L/1.5L packaged (PET) water. This is Division 1 & 2 from the original requirements document.
- Badana Industries (the Blowing Machine business) — makes preform-based bottles for three client companies: Aqua Sphere itself, Deosani, and Pivrine. This is Division 3.

Right after logging in, every role (Owner, Admin, PM, Accountant, Marketing Manager) is asked one question — which business they want to work in — and everything downstream is separate from that point on:

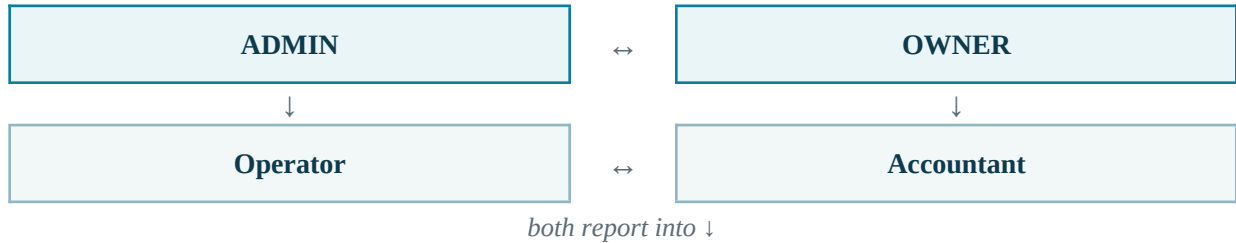


Note: These two sides never share data. Inventory, orders, and reports for the water business stay fully separate from the Blowing Machine business, even though the same five roles operate both.

1. Role Hierarchy

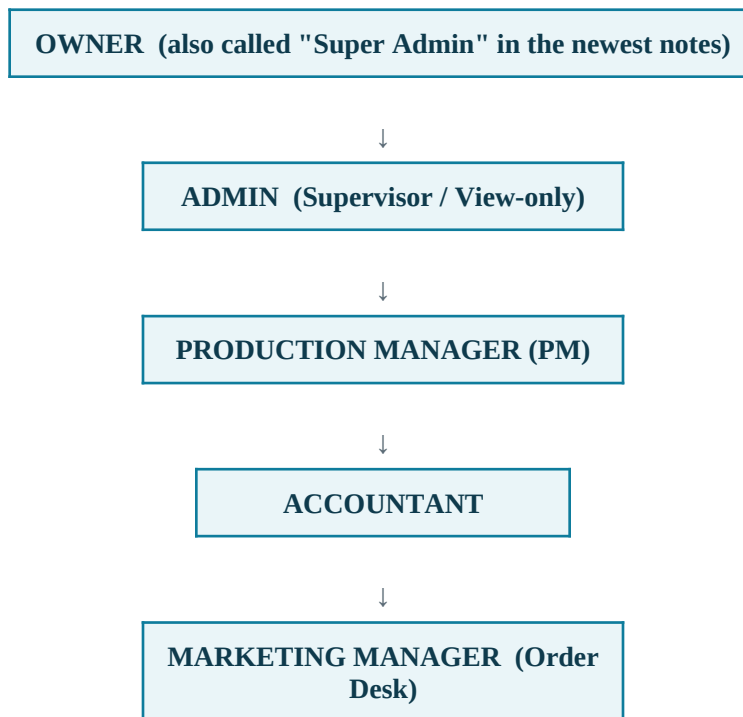
After the prototype stage, the system uses two parallel supervisory tracks that both feed into one shared operational stack.

1.1 Supervisory Split



1.2 Operational Access Stack

This is the order the system's modules are effectively "owned" in, top to bottom, across both businesses:



Note: This adds three roles that weren't in the original two-role model (Operator/Accountant vs Owner): Production Manager, Marketing Manager, and Owner acting as "Super Admin." Section 9 explains exactly what needs to change in the planning documents because of this.

Correction: The newest notes use the label "Super Admin" for a role that has the exact same powers described earlier for "Owner" — editing inventory, editing credit limits, and so on. This notebook treats Super Admin and Owner as the same person/role. If your manager actually means a fourth, separate tier above Owner, that needs to be confirmed — as written, everything points to them being the same role.

2. Role Permissions — Aquasphere (Water Business)

2.1 Admin — Supervisor (View-Only)

- Role: Supervisor. Cannot perform any task — the entire portal is view-only for this role.
- Can see: Inventory, Daily Production, Daily Orders.
- Can also see customer-related alerts (for example, a credit-limit-exceeded notice) — view-only, cannot act on them.
- End-of-day action: Admin closes the portal for the day by clicking OK — a final review-and-confirm step, not a data-entry step.

How Admin Closes the Day

Check Stock Number

→ OK

Check Production (0.5L, 1.5L)

→ OK

Check Orders

- Admin checks orders two ways: through WhatsApp, and by viewing the portal.
- Checks all orders and cash.

→ Only Supervision ←

Things Admin Cannot See

- Profit
- Cost
- Cannot place an order

By the end of the day, Admin only cross-verifies the day's data and clicks to confirm — no editing.

2.2 Owner / Super Admin

- Sees everything: daily cash, expenses, profit, losses, charts, inventory, orders → sales and profit.

The following can be seen AND edited only by the Owner:

- Overall profit and reports
- Delete a customer
- Edit Inventory option — reserved for critical situations, when records don't match reality
- All passwords — set and change
- Changes to the website
- The credit limit applied to a customer

- A customer's last-purchase record
- If a purchase is made, the Owner can see and correct exactly which inventory items it used

***Note:** The last three items above come from the newest notes, written under the label "Super Admin." They're added here to the Owner section since — as explained in Section 1.2 — this appears to be the same role.*

2.3 Production Manager (PM) — Water Business

- Enters how many 0.5L and 1.5L packs ("pats") were made; PM sees overall production.
- Records the 0.5L and 1.5L bottles that were broken during production.

What PM can see:

- Inventory → Minerals, Caps, Empty Bottles, Labels, Shrink Wraps, Already-Made PETs
- Closes their section by the end of the day

2.4 Marketing Manager (Order Desk) — Water Business

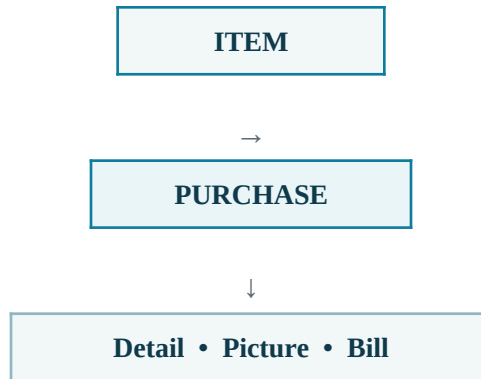
ORDER

0.5L | 1.5L | 19L

- Enters the order; enters a new customer if one doesn't already exist.
- Updates the delivery status.
- Sets the price.
- Checks whether cash is paid or not.
- Checks which method the cash was paid by.
- Can add and view customers, but cannot delete them.
- Places orders against the existing (old) customer list.
- Closes and checks orders by date.
- Checks whether today's orders are complete or not.

Note: If an order placed today is not cleared today, it stays visible to Admin and Owner, and must be completed the same day.

2.5 Accountant — Water Business



Daily Expense

- A picture is attached for every expense entry (for example, a fuel receipt photo instead of just typing "fuel").

Spot Sale (Counter Sale)

- Customers bring their own containers of different sizes, measured in litres.
- Another employee can see the sales, but everything is ultimately reviewed by the Accountant.
- Accountant records: litres of water sold, caps used, cash collected, total credit.

Other Accountant Actions

- Alternative entry method: attach a picture of the paper ledger, then add typed details.
- Cross-verifies all entries.
- Records how much cash was collected and produces a report.
- Generates an invoice.

What the Accountant Can Access

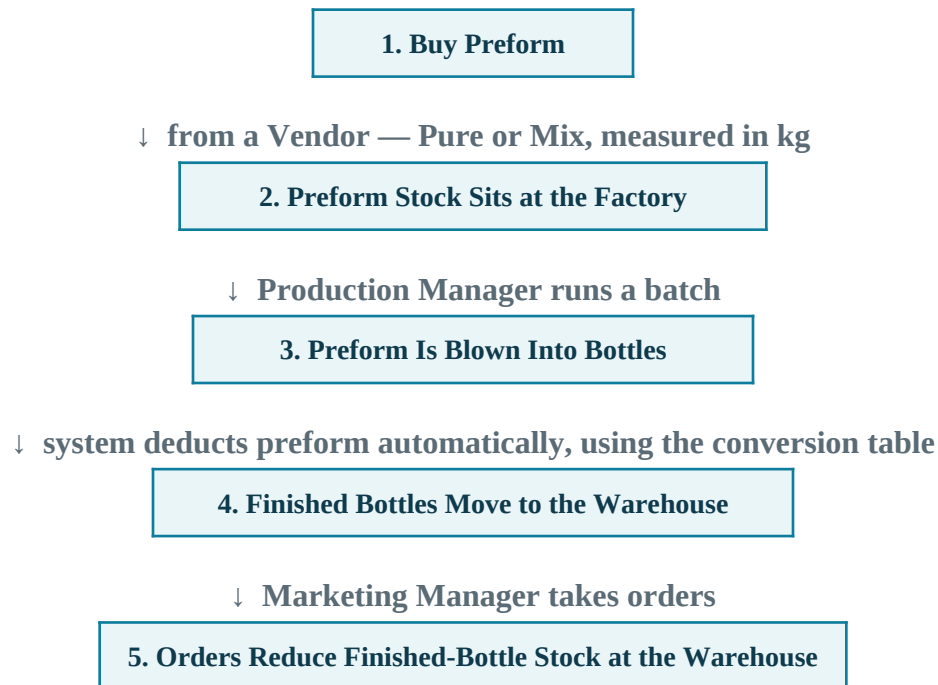
- Can see customers.
- Has no concern with (no access to) Inventory.

3. Division 3 — Badana Industries (Blowing Machine)

This division makes preform-based bottles for three client companies. It runs completely separately from the water business (Section 0), but uses the same five roles. This section is expanded in full detail below — every note from the manager's third page of notes is covered, with a plain-language explanation of what it means to actually build, since this division has the most new, previously-undocumented logic.

3.0 How This Division Works, Start to Finish

Before the detailed rules, here is the whole cycle in plain terms, so every rule below has a place to sit:



Note: This 5-step cycle is implied across several separate note pages — it isn't written this way in the original notes, but every rule below only makes sense in light of it, so it's laid out first here for clarity.

3.1 Client Companies & Preform Type

Badana Industries makes bottles for three separate client companies. Each company only ever uses one or both of two raw preform types — "Pure" and "Mix" — and the system must always keep these two types as separate, distinct inventory quantities, never combined into one number.

Correction: The handwriting first wrote the second company's name as "Dasai," then corrected itself to "Deosani." The manager's own note calls this out explicitly as "a mistake in documentation" — meaning the earlier spelling "Deosai" (used in the very first project blueprint) is likely the error, and "Deosani" is the correct spelling going forward. Confirm once, then use "Deosani" consistently everywhere: code, database, and every document.

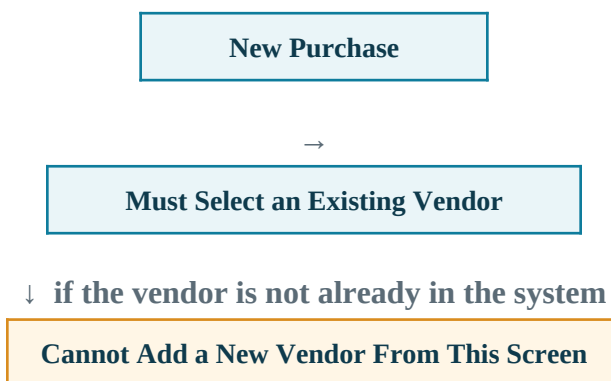
Company	Preform Type(s) Used	What This Means
Aqua Sphere	Pure only	The company's own bottles — only ever draws from Pure preform stock.
Deosani (see correction above)	Pure + Mix	A client company — can draw from either preform type.

Pivrifine	Pure + Mix	A client company — can draw from either preform type.
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- The Marketing Manager must be able to check and filter orders separately per company — three independent order lists, not one shared list.
- Because Aqua Sphere only ever uses Pure preform, the order/production form for Aqua Sphere doesn't need to ask which preform type to use — Deosani and Pivrifine do, since they use both.

3.2 Purchasing & Vendor Rule (Applies to the Whole System, Not Just Blowing)

***Note:** This rule appeared in the notes right before the Blowing Machine section, but it is a general purchasing rule — it applies to buying raw materials for the water business too, not only preform for Blowing Machine. It's kept here because that's where it was written, but it should be built once, shared across both divisions.*



Plainly: a purchase can only be recorded against a vendor that already exists in the system. There is no "quick add" shortcut to create a vendor halfway through filling out a purchase — the purchase form simply cannot be submitted without a valid, existing vendor attached to it.

- Solution the manager specified: add a dedicated "Add New Vendor" button and form, completely separate from the purchase form.
- The correct order of operations is therefore: (1) if the vendor is new, go set them up first via "Add New Vendor," then (2) come back to the purchase form and select them from the now-updated vendor list.
- For a developer: the Purchase form's vendor field should be a searchable dropdown of existing vendors only — never a free-text field, and never a field with an inline "create new" option.

3.3 Inventory Locations — Factory vs. Warehouse

Blowing Machine inventory is physically split across two locations, and the system must track stock at each location separately, not as one combined number:

Location	What's Stored There
Factory	Raw preform stock — both Pure and Mix, measured in kg. This is material that hasn't been blown into a bottle yet.
Warehouse	Finished bottles — already blown, ready to be sold/delivered against an order.

Why this matters: the business needs to answer two very different questions at any moment — "how much raw preform do we have left to work with?" (Factory) and "how many finished bottles do we actually have to sell?" (Warehouse). Combining these into one number would hide both answers.

Worked example, exactly as the manager wrote it for Deosani:

(Pure + Mix) Deosani

- The Production Manager enters, for every single batch, whether the preform used came from the Factory or the Warehouse — this is what keeps both locations' numbers accurate over time (see 3.6).

Please confirm with manager: It's a little unusual for finished preform-turned-bottles to ever be recorded as coming "from the Warehouse" as an input to production, since the Warehouse is described as holding finished bottles, not raw preform. This might mean PM is choosing which location's raw-material count to deduct from in cases where some raw preform is also kept at the Warehouse as overflow. Worth confirming the exact reason preform could be drawn from either location before building the production form.

3.4 Preform — Exact Conversion

These figures are measured in kilograms of preform needed to make one single bottle, and they are the numbers the system uses to automatically deduct raw material stock every time a production batch is entered:

Bottle Size	Preform Type	Preform Weight (per bottle)
0.5L	Mix	27 g
1.5L	Mix	15 g
1.5L	Pure	13 g
0.5L	Pure	15 g

Please confirm with manager: As written, the smaller 0.5L Mix bottle (27g) uses more preform than the larger 1.5L Mix bottle (15g) — the opposite of what you'd normally expect from bottle size. It's transcribed exactly as written, but it's worth a quick double-check with the manager before it goes into the production formulas, since a swap here would make every Mix-preform deduction wrong in the wrong direction.

How this is used — worked example:

If the Production Manager records...	The system should automatically deduct...
500 bottles of 1.5L Mix produced	$500 \times 15\text{g} = 7,500\text{g} = 7.5 \text{ kg}$ from Mix preform stock
200 bottles of 0.5L Pure produced	$200 \times 15\text{g} = 3,000\text{g} = 3.0 \text{ kg}$ from Pure preform stock

- The deduction always happens from whichever location (Factory or Warehouse) the Production Manager selected for that batch — see Section 3.3.
- Pure and Mix stock must be tracked and deducted as two entirely separate running totals — never merged into a single "preform" number.

3.5 Add Customer — For Future Companies

This is a separate, forward-looking request from the manager, distinct from the three named companies above.

- Add an "Add Customer" (or "Add Company") option to the Blowing Machine module.
- Reason, in the manager's own words: new companies may approach AQUA Sphere in the future and ask to have bottles built for them — the system should be ready to onboard a brand-new client company through the UI, without needing a developer to add hardcoded support for it.
- Practically: Aqua Sphere, Deosani, and Pivrefine should not be hardcoded as the only three possible companies — they should be the first three rows in a "Companies" table that new entries can be added to later.

3.6 Production Manager (PM) — Blowing Machine Duties

- Enters each production batch: bottle size (0.5L/1.5L), preform type used (Pure/Mix), and quantity of bottles made.
- The system automatically decreases the correct preform quantity, using the exact conversion table in Section 3.4, the moment a batch is entered — PM never does this math by hand.
- For every batch, PM also enters which location the preform was drawn from — Factory or Warehouse (Section 3.3) — so both locations' stock numbers stay accurate.
- This mirrors the same principle used everywhere else in this system: PM enters simple, observable facts (what was made, how much, from where); the system does all the derived math.

3.7 Marketing Manager — Blowing Machine Duties

- Checks and manages orders separately for each of the three client companies: Aqua Sphere, Deosani, Pivirifine — three distinct, filterable order views.
- Also looks up current Inventory levels (both Factory preform stock and Warehouse finished-bottle stock) before confirming an order can be fulfilled.

3.8 Accountant — Blowing Machine Duties

Tracks every operating expense specific to running the Blowing Machine:

- Electricity bills
- Workers' salaries
- Preform cost (what was paid to vendors for raw Pure/Mix preform)
- Machine maintenance / upgrade cost, if and when that's required

As with the water business (Section 2.5), these expenses affect profit reporting only — they never directly touch inventory quantities.

3.9 Admin — Blowing Machine Duties

- Same view-only, supervisory role as in the water business (Section 2.1).
- Can see customers/companies that are currently flagged by a system alert (for example, a company approaching or exceeding a credit limit) — this is a filtered, alert-driven view, not the full company list.
- Cannot place orders, cannot edit anything, and has no visibility into profit or cost — identical restrictions to the water-business Admin role.

3.10 Owner / Super Admin — Blowing Machine Duties

These are the elevated, edit-capable permissions the manager listed under "Super Admin" — as explained in Section 1.2, this notebook treats Super Admin as the same role as Owner, just documented from the Blowing Machine side:

- Can turn on an "Edit Inventory" option, for correcting Blowing Machine stock records when they don't match physical reality — the same critical-condition-only philosophy as the water business's Owner-only inventory edit (Section 2.2).
- When a purchase is recorded, Owner can see — and correct if necessary — exactly which inventory items and quantities that purchase affected.
- Can edit a company's credit limit, view their last-purchase record, and edit Inventory directly.
- Can always view Pure preform and Mix preform as two separate, distinct quantities — this permission exists specifically to guarantee the two types are never silently combined into one number, anywhere in the system.

Note: Technical note for the build: even Owner's "Edit Inventory" here should be implemented as a special, logged adjustment transaction — not a raw field edit — to stay consistent with the core system rule that no stock number is ever

*directly overwritten (see rules.md §1 and architecture.md §4). The permission is about *who* can trigger a correction, not an exception to *how* corrections are recorded.*

3.11 Full Worked Example — End to End

Putting Sections 3.1–3.10 together into one concrete walkthrough, the way a developer would trace it through the system:

Step	What Happens
1. Purchase	Accountant records a purchase of 50kg Mix preform from an existing vendor ("ABC Plastics"). Factory Mix-preform stock increases by 50kg.
2. Production	PM enters a batch: 1,000 bottles of 1.5L Mix produced for Deosani, drawn from the Factory.
3. Automatic deduction	System calculates $1,000 \times 15\text{g} = 15,000\text{g} = 15\text{kg}$, and reduces Factory Mix-preform stock from 50kg to 35kg.
4. Transfer to Warehouse	The 1,000 finished 1.5L Deosani bottles are now reflected in Warehouse finished-goods stock.
5. Order	Marketing Manager places an order for Deosani for 400 of those bottles; Warehouse stock drops from 1,000 to 600.
6. Dashboard	Owner can see, at any time: 35kg Mix preform at Factory, 600 finished bottles at Warehouse, and the expense of the original 50kg purchase reflected in profit.

4. Inventory & Mineral Calculations — Water Business

4.1 Label Weight per Pack

Bottle Size	Label Weight per Pack
0.5L → 1 Pat	6.72 g labels
1.5L → 1 Pat	7.86 g labels

4.2 Minerals

Inventory must also track Sodium, Calcium, and Magnesium, each with its required quantity, measured in kilograms.

Confirmed exact conversion:

Input	Result
2 kg Calcium	15.140 L water treated
0.5 kg Sodium	used per set

***Note:** This confirms 15,140 L (per the manager's handwritten note), not the 13,248 L figure that appeared earlier — project-requirements.md should be updated to lock in 15.140 L per 2kg Calcium as the mineral-set conversion basis, and remove this as an open item.*

4.3 Stock Alert Thresholds

Mineral	Reorder Alert Level
Sodium	3 kg
Calcium	10 kg
Magnesium	5 kg

4.4 Bottle Returns



- The "broken" option stays exactly as already defined (a good vs. broken split on return).

4.5 Raw Material — No Re-Order Needed

- In Inventory → Raw Material: no re-order feature is needed here (raw material is tracked, but reordering logic isn't required for it).

In Raw Material → Production:

Input	Result
0.5L → 1 Pat	9 L water
Labels	not yet accurate — needs confirmation

5. Credit Limit & Digital Bill Alert — Water Business

Manager's question: if a customer doesn't pay and clear their balance, and their credit limit is exceeded, can the system generate a digital bill to inform them?

Worked example:

Step	Value
Credit limit	5,000
Customer doesn't pay for a month	—
Credit exceeds to	6,000
Action	Inform the customer via WhatsApp or email

***Note:** This is a concrete example of the soft-block credit warning already defined in the requirements document — it confirms the trigger should also send an outbound WhatsApp/email notice, which wasn't specified before. This should be added as a notification requirement.*

6. Order & Customer Module — Additional Notes

In Order

- There is only a search option — no separate browse/list-first view.
- Add a button and form for a new customer order.
- Smart approach: prompt to add a new customer automatically when a new order comes in for someone not yet in the system.

In Customer

- Add a location (map pin) and a picture of the customer's house.

7. 19L Bottle Ledger — Execution Flow

This is a bottle transaction ledger, active while a transaction is executing.



When a 19L bottle transaction happens, its status must be updated in both the Purchase section and the Inventory at the same time — never one without the other.

8. Corrections & Open Questions — Full List

Everything flagged throughout this notebook, gathered in one place, so it's easy to run past the manager before building against it:

Corrections Made (handwriting fixed, logic unchanged)

- "Dasai" → "Deosani" (client company name) — but see the open question below, since the original blueprint spelled it "Deosai."
- Fragmented notes on Admin's alert visibility, PM's warehouse/factory entry, and the Super Admin permissions list were written out as full sentences.

Please Confirm With the Manager

- Final spelling of the second Blowing Machine client company: "Deosai" (original blueprint) or "Deosani" (latest notes)? Pick one and use it everywhere.
- Preform weights: the notes show 0.5L Mix using more preform (27g) than 1.5L Mix (15g) — the reverse of what's normally expected. Worth a quick verification before it's built into the production math.
- Is "Super Admin" meant to be the same person/role as "Owner," or a distinct fourth tier above Owner? This notebook assumes they're the same.
- Label conversion for the water business production formula is still marked "not accurate" in the original notes — a real figure is still needed.
- Why would a Production Manager ever draw raw preform from the Warehouse (which is described as holding finished bottles, not raw material)? Worth a quick clarification before the production form is built.

New Information That Should Update the Planning Documents

- Two new roles — Production Manager and Marketing Manager — need to be added to the role/permission model in rules.md and architecture.md, alongside Operator/Accountant/Owner.
- Admin is fully view-only/supervisory — a distinct third permission level, not folded into Operator.
- The mineral conversion open item is now resolved: 2kg Calcium → 15.140 L water.
- A credit-limit breach should trigger an outbound WhatsApp/email notice — a new notification requirement.
- Division 3 (Blowing Machine / Badana Industries) now has real detail to build from: 3 client companies, Pure/Mix preform types, Factory vs. Warehouse inventory locations, exact preform-weight conversions, and a vendor-must-exist-first purchase rule.
- Purchases across both businesses require an existing vendor — there is no create-vendor-inline shortcut. This is a purchasing-flow rule for rules.md.

Note: Recommend updating project-requirements.md, architecture.md, rules.md, and phases.md with all of the above before development reaches the roles/permissions phase and the Blowing Machine phase — just say the word and this can be folded into those files directly.